



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
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Hyperbolic Functions
Practice Problems

Formulas:

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

1. $y = x^2 \sinh(3x)$

2. $y = (2x + 1) \cosh(x^2)$

3. $y = x^3 \tanh(2x - 1)$

4. $y = (x^2 + 1) \coth(3x)$

5. $y = x^2 \operatorname{sech}(2x + 1)$

6. $y = (x - 1)^2 \operatorname{csch}(4x)$

7. $y = \sinh(2x) \cosh(3x)$

8. $y = \tanh(x^2) \operatorname{sech}(2x)$

9. $y = \frac{\cosh(2x)}{\sinh x}$

10. $y = \frac{\coth(3x-1)}{x^2+1}$



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Hyperbolic Functions
Step-by-step Solution

1. $y = x^2 \sinh(3x)$

- **Step 1:** Use the product rule.

$$\frac{d}{dx} uv = vdu + u dv$$

$$\frac{d}{dx} x^2 \sinh(3x) = 2x \sinh(3x) + x^2 (3 \cosh(3x))$$

- **Step 2:** Combine and simplify.

Answer: $y' = 3x^2 \cosh(3x) + 2x \sinh(3x)$

2. $y = (2x + 1) \cosh(x^2)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \cosh(x^2) = 2x \sinh(x^2)$$

- **Step 2:** Apply the product rule.

$$\frac{d}{dx} uv = vdu + u dv$$

$$\frac{d}{dx} (2x + 1) \cosh(x^2) = 2 \cosh(x^2) + 2x(2x + 1) \sinh(x^2)$$

Answer: $y' = 2 \cosh(x^2) + 2x(2x + 1) \sinh(x^2)$

3. $y = x^3 \tanh(2x - 1)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \tanh(2x - 1) = 2 \sec^2(2x - 1)$$

- **Step 2:** Apply the product rule.



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$$\frac{d}{dx}uv = vdu + udv$$

$$\frac{d}{dx}x^3 \tanh(2x - 1) = 3x^2 \tanh(2x - 1) + 2x^3 \operatorname{sech}^2(2x - 1)$$

Answer: $y' = 3x^2 \tanh(2x - 1) + 2x^3 \operatorname{sech}^2(2x - 1)$

4. $y = (x^2 + 1)\coth(3x)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx}\coth(3x) = -3\operatorname{csch}^2(3x)$$

- **Step 2:** Apply the product rule.

$$\frac{d}{dx}uv = vdu + udv$$

$$\frac{d}{dx}(x^2 + 1)\coth(3x) = 2x \coth(3x) + 3(x^2 - 1)\operatorname{csch}^2(3x)$$

Answer: $y' = 2x \coth(3x) + 3(x^2 - 1)\operatorname{csch}^2(3x)$

5. $y = x^2 \operatorname{sech}(2x + 1)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx}\operatorname{sech}(2x + 1) = -2\operatorname{sech}(2x + 1)\tanh(2x + 1)$$

- **Step 2:** Apply the product rule.

$$\frac{d}{dx}uv = vdu + udv$$

$$\frac{d}{dx}x^2 \operatorname{sech}(2x + 1) = 2x \operatorname{sech}(2x + 1) + -2x^2 \operatorname{sech}(2x + 1)\tanh(2x + 1)$$

Answer: $y' = 2x \operatorname{sech}(2x + 1) + -2x^2 \operatorname{sech}(2x + 1)\tanh(2x + 1)$



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6. $y = (x - 1)^2 \operatorname{csch}(4x)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \operatorname{csch}(4x) = -4 \operatorname{csch}(4x) \operatorname{coth}(4x)$$

- **Step 2:** Apply the product rule.

$$\frac{d}{dx} uv = vdu + u dv$$

$$\frac{d}{dx} x^2 (x - 1)^2 \operatorname{csch}(4x) = 2(x - 1) \operatorname{csch}(4x) - 4(x - 1)^2 \operatorname{csch}(4x) \operatorname{coth}(4x)$$

Answer: $y' = 2(x - 1) \operatorname{csch}(4x) - 4(x - 1)^2 \operatorname{csch}(4x) \operatorname{coth}(4x)$

7. $y = \sinh(2x) \cosh(3x)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \sinh(2x) = 2 \cosh(2x)$$

$$\frac{d}{dx} \cosh(3x) = 3 \sinh(3x)$$

- **Step 2:** Apply the product rule.

$$\frac{d}{dx} uv = vdu + u dv$$

$$\frac{d}{dx} \sinh(2x) \cosh(3x) = (2 \cosh(2x)) \cosh(3x) + \sinh(2x) (3 \sinh(3x))$$

Answer: $y' = 2 \cosh(2x) \cosh(3x) + 3 \sinh(2x) \sinh(3x)$

8. $y = \tanh(x^2) \operatorname{sech}(2x)$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \operatorname{sech}(2x) = -2 \operatorname{sech}(2x) \tanh(2x)$$



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$$\frac{d}{dx} \tanh(x^2) = 2\operatorname{sech}^2(x^2)$$

- **Step 2:** Apply the product rule.

$$\frac{d}{dx} uv = vdu + u dv$$

$$\frac{d}{dx} \tanh(x^2) \operatorname{sech}(2x) = 2\operatorname{sech}^2(x^2) \operatorname{sech}(2x) - 2\tanh(x^2) \operatorname{sech}(2x) \tanh(2x)$$

Answer: $y' = 2\operatorname{sech}^2(x^2) \operatorname{sech}(2x) - 2\tanh(x^2) \operatorname{sech}(2x) \tanh(2x)$

9. $y = \frac{\cosh(2x)}{\sinh x}$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \cosh(2x) = 2\sinh(2x)$$

- **Step 2:** Apply the quotient rule.

$$\frac{d}{dx} \frac{u}{v} = \frac{vdu - u dv}{v^2}$$

$$\frac{d}{dx} \frac{\cosh(2x)}{\sinh x} = \frac{\sinh x (2\sinh(2x)) - \cosh(2x) (\cosh x)}{\sinh^2 x}$$

Answer: $y' = \frac{2 \sinh(x) \sinh(2x) - \cosh(2x) \cosh(x)}{\sinh^2 x}$

10. $y = \frac{\coth(3x-1)}{x^2+1}$

- **Step 1:** Apply the chain rule.

$$\frac{d}{dx} \coth(3x - 1) = -3\operatorname{csch}^2(3x - 1)$$

- **Step 2:** Apply the quotient rule.

$$\frac{d}{dx} \frac{u}{v} = \frac{vdu - u dv}{v^2}$$



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$$\frac{d}{dx} \frac{\coth(3x-1)}{x^2+1} = \frac{-3(x^2+1)\operatorname{csch}^2(3x-1) - \coth(3x-1)(2x)}{(x^2+1)^2}$$

Answer: $y' = \frac{-3(x^2+1)\operatorname{csch}^2(3x-1) - \coth(3x-1)(2x)}{(x^2+1)^2}$